

Detection of Adulterants in Milk Using SPR Sensor

A Project Report

submitted by

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in partial fulfilment of requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY



**Department of Electronics and Communication Engineering
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING KANCHEEPURAM**

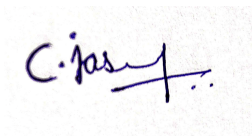
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I, **Jaswanth Reddy Chinnavula**, with Roll No: **EC21B1073** hereby declare that the material presented in the Project Report titled **Detection of Adulterants in Milk Using SPR Sensor** represents original work carried out by me in the **Department of Electronics and Communication Engineering** at the Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram.

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Jaswanth Reddy Chinnavula

Place: Chennai

Date: 08.05.2025

CERTIFICATE

This is to certify that the report titled **Detection of Adulterants in Milk Using SPR Sensor**, submitted by **Jaswanth Reddy Chinnavula (EC21B1073)**, to the Indian Institute of Information Technology, Design and Manufacturing Kancheepuram, for the award of the degree of **BACHELOR OF TECHNOLOGY** is a bona fide record of the work done by him/her under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my project guide, Dr. Srijith K, Assistant Professor in the Department of Electronics and Communication Engineering at IIITDM Kancheepuram, for his invaluable guidance and unwavering support throughout this project. His expertise, advice, and encouragement were instrumental in the successful completion of this work.

I am deeply thankful to the PhD scholars, Archana and Vanniya, for their constant assistance and insightful suggestions during the course of this project. Their contributions were crucial in overcoming challenges and refining the outcomes.

Lastly, I am grateful to my friends for their constant support, motivation, and encouragement throughout this journey. Their presence made this experience not only productive but also enjoyable.

ABSTRACT

This project report presents the design and analysis of a Surface Plasmon Resonance (SPR) sensor for detecting milk adulterants, specifically melamine and fat. The proposed sensor utilizes a multilayer configuration comprising BK7–Cr–Ag–BP–Ta₂O₅, optimized through the Transfer Matrix Method (TMM) to achieve high sensitivity and accuracy. The study systematically investigates the impact of layer thicknesses (Ag: 50 nm, BP: 0.6 nm, Ta₂O₅: 13 nm) on sensor performance, demonstrating superior sensitivity (259.82°/RIU), Figure of Merit (FOM: 62.34), and Detection Accuracy (DA: 0.24036) compared to existing techniques. The sensor's ability to detect refractive index changes in the range of 1.325–1.365 makes it highly effective for identifying adulterants in milk. Real-time, label-free detection capabilities and minimal sample preparation highlight the sensor's potential as a reliable tool for food safety applications.

Keywords: Surface Plasmon Resonance (SPR), Transfer Matrix Method (TMM), Milk Adulteration, Melamine Detection, Fat Detection.

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ABBREVIATIONS

ATR	Attenuated Total Reflection
ELISA	Enzyme-Linked Immunosorbent Assay
ESI	Electrospray Ionization
LC	Liquid Chromatography
LOD	Limit of Detection
MIPs	Molecularly Imprinted Polymers
MS	Mass Spectrometry
m/z	Mass-to-Charge Ratio
RIU	Refractive Index Unit
SPR	Surface Plasmon Resonance
TMM	Transfer Matrix Method
UV	Ultraviolet

NOTATION

λ	Wavelength
d_s	Thickness of the s -th layer
ε_s	Dielectric constant of the s -th layer
n_1	Refractive index of the incident medium
θ_1	Angle of incidence in the first layer
μ_s	Permeability of the s -th layer
q_s	Characteristic admittance of the s -th layer
β_s	Phase thickness of the s -th layer
M_{ij}	Elements of the transfer matrix
R_p	Reflection coefficient for p-polarized light
q_1	Characteristic admittance of the incident medium
q_N	Characteristic admittance of the last layer
$\Delta\theta_{res}$	Change in resonance angle
Δn	Change in refractive index
S_θ	Sensitivity in degrees per refractive index unit (RIU)

CHAPTER 1

INTRODUCTION

1.1 Background

1.1.1 Importance of Milk

Milk is a versatile and essential part of daily life for many people worldwide. Its nutritional benefits and wide range of uses in cooking, beverages, and cultural practices make it a fundamental food item in households across the globe. Whether consumed directly, used in recipes, or transformed into other dairy products, milk continues to play a vital role in nutrition and daily routines.

1.1.2 What is Melamine?

Melamine is a white, crystal-like solid with the chemical formula $C_3H_6N_6$. It has a lot of nitrogen and is mainly used to make plastics, adhesives, coatings, laminates, and tableware like melamine dishes. Because it has so much nitrogen, some people have illegally added it to food, like milk, to make it seem like it has more protein. This works because protein tests check for nitrogen, and melamine can trick the test into showing a higher protein level than there really is [1].

1.2 Motivation

Milk is a staple food in most of the household because of its health benefits. Milk provides proteins and calcium which is necessary to build muscle, teeth and bones. It's packed with important nutrients like phosphorus, B vitamins, potassium and vitamin D. It is excellent source of protein. The increasing awareness among the people to consume

protein rich foods after covid. This makes milk has important addition to most of the people's diet. Due to limited consumer awareness regarding high-quality milk products. The milk based companies commercialized into different milk products like high protein milk and adding different adulterations to increase the false protein content to make monetary profit. There exist multiple methods for incorporating adulterants into milk, hence degrading the overall quality of the product. The consumption of adulterated milk detrimental effects on our overall well being and poses a significant risk to our health.

There are several methods for detection of adulterants. The methods include liquid chromatography, mass spectrometry, enzyme-linked immunosorbent assay, and various methods too. Liquid Chromatography is a technique used to separate components of a mixture based on their interactions with a stationary phase and a mobile phase. Mass Spectrometry is used to identify and quantify compounds based on their mass-to charge ratio . However, these techniques have some limitations. Liquid Chromatography technique be time-consuming and requires skilled operator. Mass Spectrometry techniques are expensive and sample preparation is complex. In order to address these issues the optical techniques have stood out because of their label-free detection, highly sensitive, and miniature in size.

The surface plasmon resonance (SPR) sensor is commonly used in optical sensing because it offers real-time, label-free detection with high sensitivity and accuracy. These sensors are used to food safety, health monitoring systems and drugs. [1].

1.3 Organization of the Report

The report is structured into five chapters as follows:

- **Chapter 1: Introduction**
Introduces the significance of milk, the issue of melamine adulteration, and the motivation for using SPR-based optical detection methods.
- **Chapter 2: Literature Survey**
Reviews conventional and modern techniques for melamine detection, with emphasis on SPR principles, advantages, and applications.
- **Chapter 3: Methodology**

Details the multilayer SPR sensor design and the use of TMM for performance modeling.

- **Chapter 4: Results and Discussion**

Presents layer optimization, error analysis, and performance evaluation for fat and melamine detection, with comparison to existing sensors.

- **Chapter 5: Conclusion**

Summarizes key findings and highlights the sensor's potential for real-time, label-free milk adulterant detection.

Acknowledgment of Online Tools

In the preparation of this chapter, ChatGPT was utilized to assist in drafting and refining the content of the following sections:

- Section 1.1: Theory about milk.
- Section 2.2.1: Language refinement and formalization.

The assistance provided by ChatGPT involved generating text, rephrasing content, and ensuring formal tone consistency.

CHAPTER 2

Literature Survey

2.1 Different Techniques of Detection of Melamine in Milk

2.1.1 Liquid Chromatography (LC)

Liquid Chromatography is a technique used to separate different parts of a mixture. It works by letting a liquid move through a material that stays in place, causing the substances in the mixture to separate based on how they interact with these materials.

To detect melamine in milk, a milk sample is injected into the LC system. Inside the system, the sample passes through a column filled with a solid or liquid material. A liquid, which is usually a solvent or a mix of solvents, carries the sample through the column. Melamine and other substances in the milk interact differently with the stationary phase. Because of these differences, they move at different speeds, allowing them to separate. After separation, special detectors, like UV absorption, are used to identify melamine based on its unique light absorption properties.[2].

Limitations of Liquid Chromatography (LC) method:

- Time-consuming separation process.
- Possible sample loss during analysis.
- Requires skilled operation.
- Limited sensitivity without advanced detectors.

2.1.2 Mass Spectrometry (MS)

Mass Spectrometry (MS) is used to identify and quantify compounds based on their mass-to charge ratio (m/z). For detecting melamine in milk, once the components are

separated by LC, they are introduced into the mass spectrometer. The sample is ionized, typically using Electrospray Ionization (ESI), converting the molecules into charged ions. The ions are then sent into the mass analyzer, where they are sorted based on their mass-to-charge (m/z) ratios. Melamine has a unique m/z ratio, which allows it to be detected and measured accurately. The resulting mass spectrum provides a detailed analysis of the ions in the sample, allowing the identification of melamine by its specific peak and its abundance relative to other compounds. This precise identification is crucial in detecting trace amounts of melamine in milk[2].

Limitations of Mass Spectrometry (MS) method:

- Expensive equipment and maintenance.
- Requires high expertise to operate and interpret data.
- Sample preparation can be complex.
- Matrix effects can interfere with detection.

2.1.3 Enzyme Linked Immunosorbent Assay(ELISA)

Enzyme Linked Immunosorbent Assay is an effective and widely used technique for detecting melamine contamination in milk, ensuring food safety. ELISA works by using a monoclonal or polyclonal anti-melamine antibody, which specifically binds to melamine molecules in the milk sample. The process starts by coating a microplate with either these anti-melamine antibodies or melamine itself (in a competitive ELISA format). When the milk sample is added, the melamine in it attaches to the antibodies that are already coated on the surface. After washing away any unbound substances, a secondary antibody, linked to an enzyme such as horseradish peroxidase (HRP), is introduced. This secondary antibody binds to the anti melamine antibody, and the enzyme reacts with a substrate to produce a detectable signal, usually a color change. The strength of the signal increases as the amount of melamine in the sample goes up. By comparing the signal to a standard curve, the exact amount of melamine can be quantified. ELISA is highly sensitive and specific, allowing it to detect even trace amounts of melamine, making it an invaluable tool for routine testing in food safety laboratories[2].

Limitations of Enzyme-Linked Immunosorbent Assay method:

- Potential interference from structurally similar compounds may affect accuracy.
- May not detect extremely high or low concentrations without proper optimization.
- Requires high-quality anti-melamine antibodies, which can be costly.
- Milk samples may need dilution or preprocessing to remove interfering substances.

2.1.4 Surface Plasmon Resonance

Surface Plasmon Resonance (SPR) sensors offer a unique combination of label-free, real-time, and highly sensitive detection capabilities, making them a preferred choice for studying molecular interactions. SPR can detect very small changes in refractive index, enabling the measurement of low analyte concentrations.

2.2 Surface Plasmon Resonance**2.2.1 Working Principle**

A Surface Plasmon Resonance sensor works by exciting surface plasmons, which are waves of free electrons moving together at the boundary between a metal and another material. When polarized light hits the metal surface at a specific angle, it interacts with these free electrons and creates surface plasmons. This interaction results in a decrease in the intensity of the reflected light, which appears as a trough in the reflectance spectrum. The resonance angle is very sensitive to refractive index changes in the vicinity of the metal surface. When ligands immobilized on the metal surface bind to target molecules, the refractive index at the interface is changed, resulting in a shift of the resonance angle. This change is sensed and quantified, giving real-time, label-free information concerning molecular interactions, such as binding kinetics, affinity, and concentration. Being able to watch these interactions dynamically and with high sensitivity gives SPR a strong capability for investigating biomolecular processes[3].

2.2.2 Applications of SPR Sensors

- **Contaminant Detection:** Detect harmful substances like melamine, pesticides, and toxins in food products.
- **Pollutant Detection:** Monitor pollutants like heavy metals, pesticides, and organic compounds in water and air.
- **Pesticide Residue Analysis:** Detect pesticide residues in crops and soil.

2.2.3 SPR detection of melamine in milk

Surface Plasmon Resonance (SPR) technology finds considerable importance in melamine detection in milk as it is extremely sensitive, possesses real-time analysis capabilities, and detects in a label-free format. Melamine, a toxic chemical illegally added to milk to artificially inflate protein content, poses serious health risks, making its detection crucial for food safety. SPR sensors can be functionalized with specific receptors, such as antibodies or molecularly imprinted polymers (MIPs), that selectively bind to melamine. When melamine-contaminated milk is introduced to the sensor, the binding of melamine to these receptors causes a change in the refractive index at the metal-dielectric interface. This is measured as a change in the SPR signal, allowing for precise and quantitative measurement of melamine concentration. SPR's real-time, label-free detection capacity renders it a quick and reliable tool for melamine adulteration identification in milk, ensuring food safety and compliance with regulatory standards[3].

2.2.4 Advantages over traditional Sensors

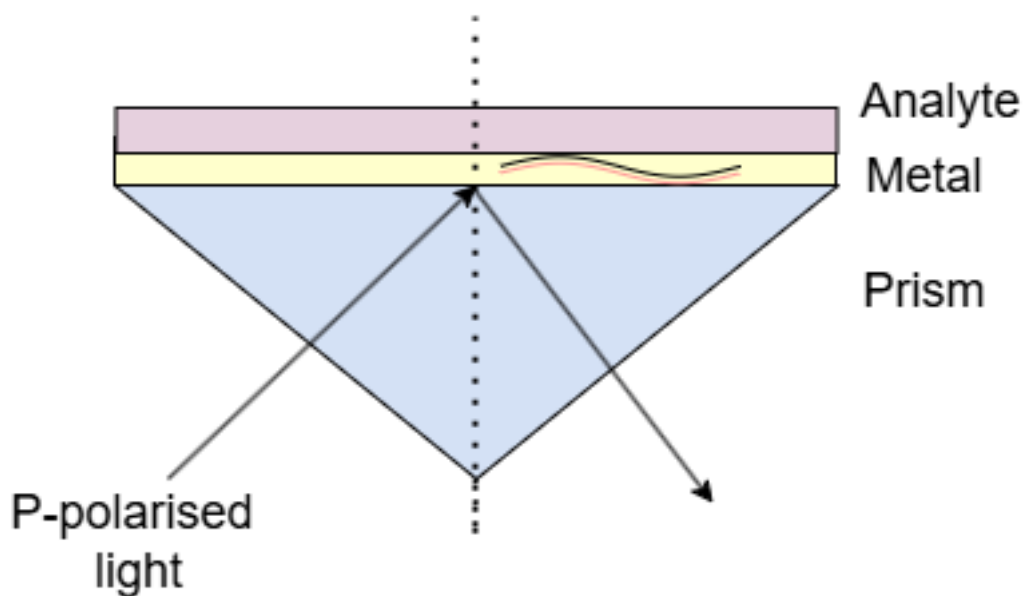
Table 2.1: Comparison between SPR and Traditional Methods

Parameter	SPR	Traditional Methods	Why SPR is Better
Sensitivity	High	High to moderate	SPR offers comparable or better sensitivity without complex sample preparation.
Speed	Real-time detection (minutes)	Time-consuming (hours)	SPR provides immediate results, enabling real-time monitoring.
Cost	Moderate initial cost, lower operational cost	High	SPR has lower long-term costs due to minimal reagent use and no labeling requirements.
Complexity	Simple operation, minimal sample preparation	Complex	SPR is user-friendly and requires less technical expertise.
Specificity	High (can be functionalized with specific receptors for melamine)	High to moderate	SPR offers high specificity through tailored receptor-analyte interactions.

2.3 Kretschmann configuration

The Kretschmann setup is the most common way to excite surface plasmons in Surface Plasmon Resonance (SPR) sensors. This setup works using a process called attenuated total reflection (ATR). It includes a prism with a thin layer of metal on one side.

When polarized light passes through the prism at a specific angle, it hits the metal-dielectric surface and excites surface plasmons. This causes a sharp drop in the reflected light intensity, which is recorded as the SPR signal. The resonance angle is very sensitive to small changes in the refractive index near the metal surface, making the Kretschmann setup ideal for detecting molecular interactions in real time [3].



Kretschmann Configuration

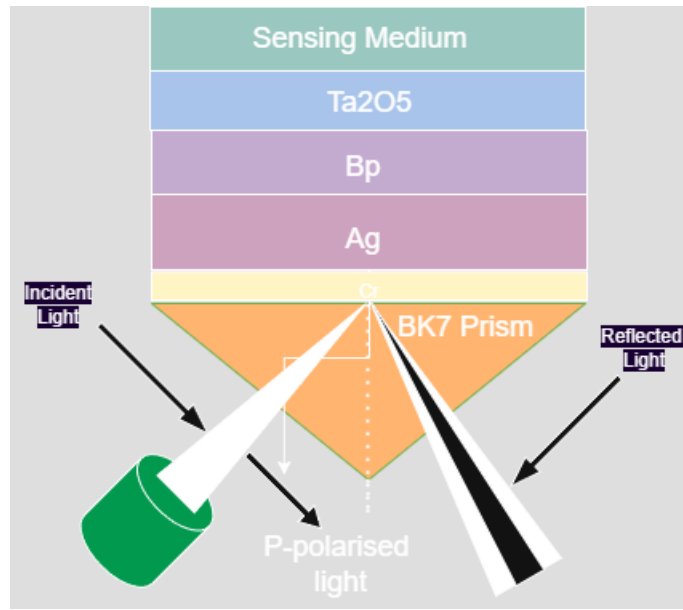
2.4 Summary

Melamine detection in milk is essential due to its toxic effects and illegal use in food adulteration. Techniques such as Liquid Chromatography (LC) and Mass Spectrometry (MS) provide accurate identification but require complex procedures, expensive equipment, and skilled operation. Enzyme-Linked Immunosorbent Assay (ELISA) offers high sensitivity and specificity using antibodies but may suffer from cross-reactivity and requires high-quality reagents. Surface Plasmon Resonance (SPR) has emerged as a promising alternative, offering label-free, real-time detection with high sensitivity and minimal sample preparation. The Kretschmann configuration is commonly employed in SPR to excite surface plasmons through attenuated total reflection. Additionally, the Transfer Matrix Method (TMM) is used to simulate and optimize multilayer SPR structures for effective melamine sensing.

CHAPTER 3

Methodology

3.1 Sensor Design



SPR Sensor

Performance of the SPR sensor is mainly dependent on the materials and their arrangement. The proposed Sensor design utilizes BK7 as substrate. A 5-nm thick chromium (Cr) film is used as an adhesion layer, which is important for practical implementation. This layer helps prevent issues such as metal etching that can occur when depositing metal directly onto a prism. Chromium also forms a stable oxide layer that protects it from further oxidation [4]. Silver (Ag) is commonly used in SPR sensors due to its high sensitivity in the ultraviolet to visible light range, while also maintaining stable performance. It is coated on the top of Cr layer. Furthermore, a layer of black phosphorus (BP) is applied over the metallic structure as field enhancement layer. An additional layer of Ta₂O₅ is applied over the black phosphorus (BP) layer to enhance the sensor's performance. Ta₂O₅ is a metal oxide with a high refractive index and contains a significant number of defects, particularly oxygen vacancies. These defects

improve the adsorption of analyte molecules by increasing their interaction with the sensing surface, leading to better coupling with the evanescent field and stronger SPR signal generation, which results in higher sensitivity. Moreover, Ta_2O_5 offers excellent chemical and thermal stability. The refractive index (RI) of the materials used is a key parameter in sensor modeling [5]. In this study, the RI of the BK7 prism is calculated using the following equation

$$n_{\text{prism}} = \left(1 + \frac{1.03961212\lambda^2}{\lambda^2 - 0.00600069867} + \frac{0.231792344\lambda^2}{\lambda^2 - 0.0200179144} + \frac{1.0104694\lambda^2}{\lambda^2 - 103.560653} \right)^{\frac{1}{2}}$$

The refractive Index of Ta_2O_5 layer is calculated as follows

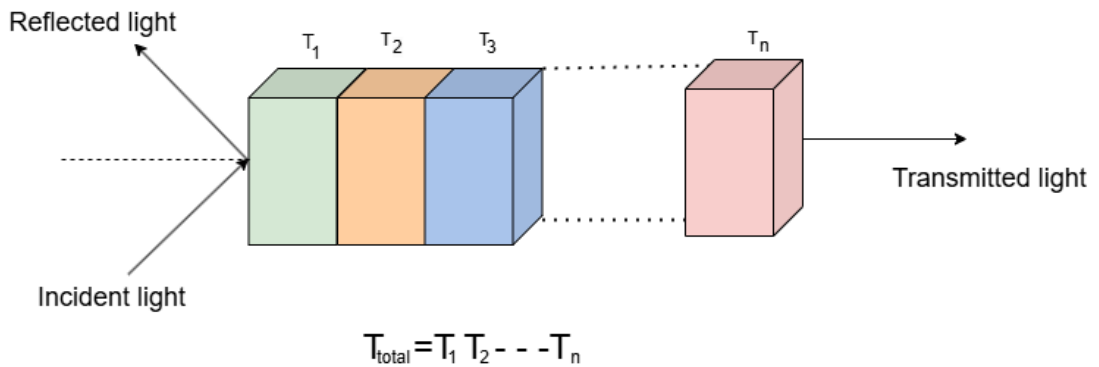
$$n_{\text{Ta}_2\text{O}_5} = 1.88 + 178.4 \times \frac{10^2}{\lambda^2} + 52.7 \times \frac{10^7}{\lambda^4}$$

where λ is the wavelength (in nm), i.e., 633 nm. Table 1 summarizes the RI values of different materials utilized in the sensor proposed at a wavelength of 633 nm.

Table 3.1: RI values of the materials at $\lambda = 633$ nm

Material	RI	Ref
Chromium (Cr)	$3.1414 + 3.3143i$	[6]
BP	$3.5 + 0.01i$	[6]
Ag	$0.056 + 4.2776i$	[4]
Ta_2O_5	$1.9278 + 0.0i$	[5]

3.2 Mathematical Model of TMM



$$M_{ij} = \left(\prod_{s=2}^{N-1} M_s \right)_{ij} = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix}$$

$$M_s = \begin{pmatrix} \cos \beta_s & -\frac{i \sin \beta_s}{q_s} \\ -iq_s \sin \beta_s & \cos \beta_s \end{pmatrix}$$

where

q_s is the characteristic admittance of the s -th layer

β_s is the Phase thickness of the s -th layer

The phase thickness of the s -th layer is given by:

$$\beta_s = \frac{2\pi d_s}{\lambda} (\varepsilon_s - n_1^2 \sin^2 \theta_1)^{1/2}$$

where:

- d_s is the thickness of the s -th layer.
- λ is the wavelength of incident light.
- ε_s is the dielectric constant of the s -th layer.
- n_1 is the refractive index of the incident medium.
- θ_1 is the incidence angle of first layer.

The characteristic admittance of the s -th layer is:

$$q_s = \sqrt{\frac{\mu_s}{\varepsilon_s}} \cos \theta_s = \frac{(\varepsilon_s - n_1^2 \sin^2 \theta_1)^{1/2}}{\varepsilon_s}$$

where:

- μ_s s -th layer permeability.
- ε_s - s -th layer dielectric constant.
- θ_1 - incidence angle .

$$R_p = \left| \frac{(M_{11} + M_{12}q_N)q_1 - (M_{21} + M_{22}q_N)}{(M_{11} + M_{12}q_N)q_1 + (M_{21} + M_{22}q_N)} \right|^2$$

where:

- R_p - reflection coefficient.
- M_{ij} - elements of the transfer matrix.
- q_1 - the characteristic admittance of the 1-st layer.
- q_N - the characteristic admittance of the N -th layer.

CHAPTER 4

Results & Discussion

4.1 Optimization of Layer Thicknesses

The refractive index of common milk adulterants typically falls within the range of 1.325 to 1.365. Based on this range, a systematic optimization of the multilayer surface plasmon resonance (SPR) sensor configuration was carried out to achieve maximum sensitivity and improved performance metrics.

The initial structure considered for this optimization process was BK7–Cr–Ag. Within this configuration, the thickness of the silver (Ag) layer was varied from 25 nm to 70 nm. This step aimed to identify a thickness that offers a favorable compromise between sensitivity and Full Width at Half Maximum (FWHM) of the resonance curve. After evaluating the trade-offs, a thickness of 50 nm for the Ag layer was selected, as it provided high sensitivity while maintaining a relatively narrow FWHM, which is essential for precise resonance angle detection.

Following this, the structure was extended to BK7–Cr–Ag–BP in order to investigate the impact of incorporating a black phosphorus (BP) layer. The BP layer thickness was varied between 0.2 nm and 1.0 nm. The optimization analysis revealed that a BP thickness of 0.6 nm yielded the best performance in terms of resonance angle shift and overall sensor response, indicating its suitability for enhancing plasmonic interactions.

Subsequently, the configuration was further expanded to include a dielectric layer, resulting in the structure BK7–Cr–Ag–BP–Ta₂O₅. The Ta₂O₅ layer was analyzed with thickness values ranging from 6 nm to 13 nm to determine its effect on sensor sensitivity. Among these, a thickness of 13 nm was identified as optimal due to its ability to significantly enhance sensitivity while maintaining a stable resonance profile.

In summary, the optimized configuration for the SPR sensor consists of the following layer thicknesses:

- Silver (Ag): 50 nm
- Black Phosphorus (BP): 0.6 nm
- Tantalum Pentoxide (Ta_2O_5): 13 nm

This optimized multilayer structure is expected to provide superior performance for the detection of milk adulterants across the typical refractive index range.

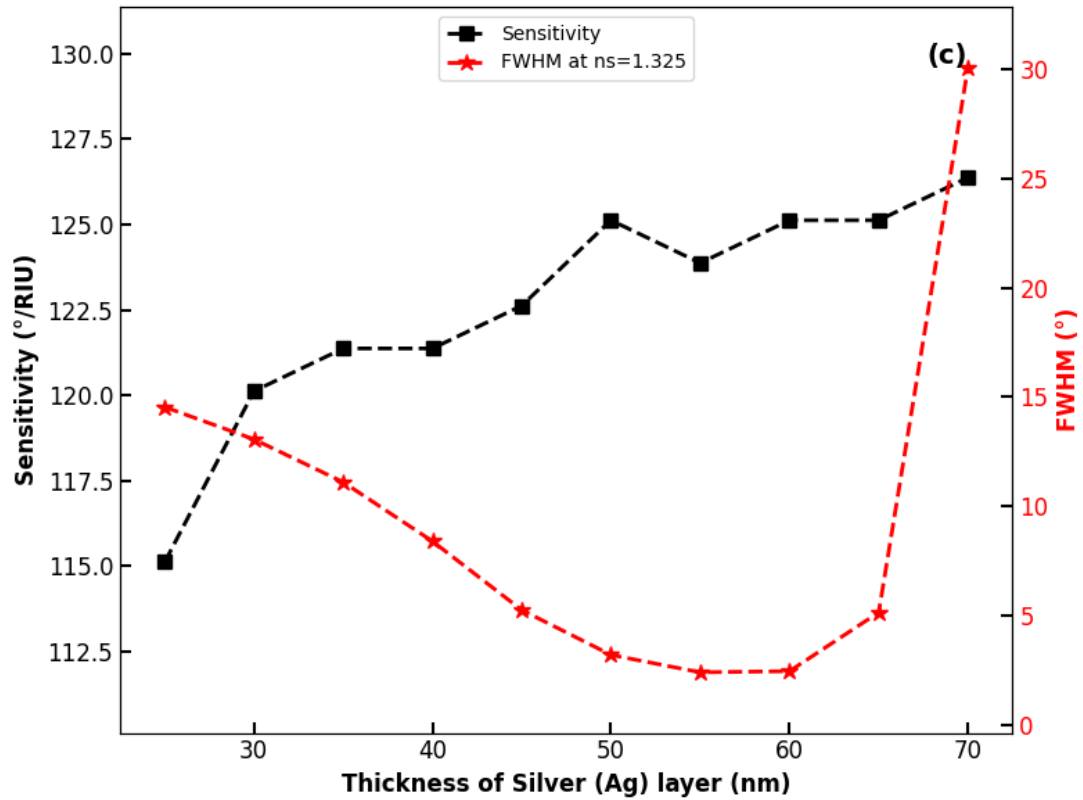


Figure 4.1: Optimization of Ag layer thickness.

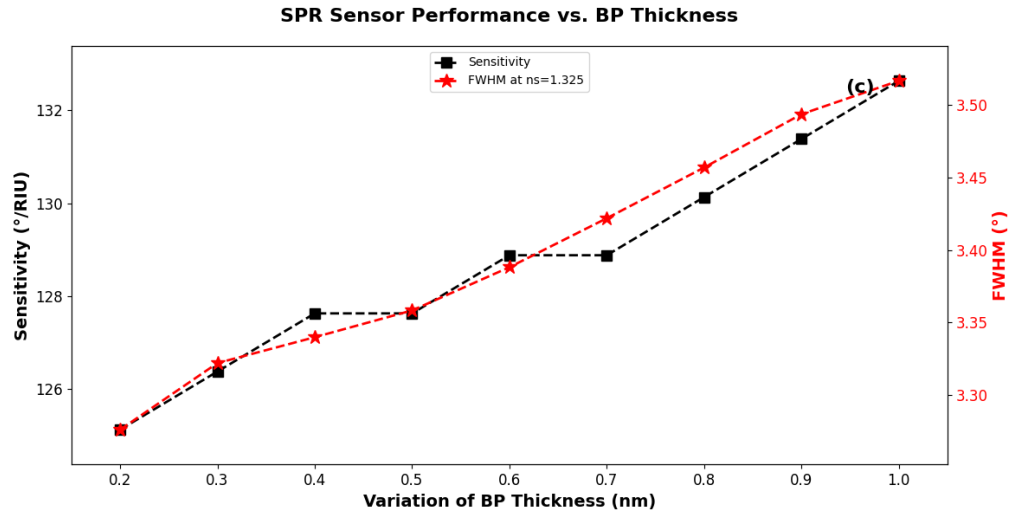
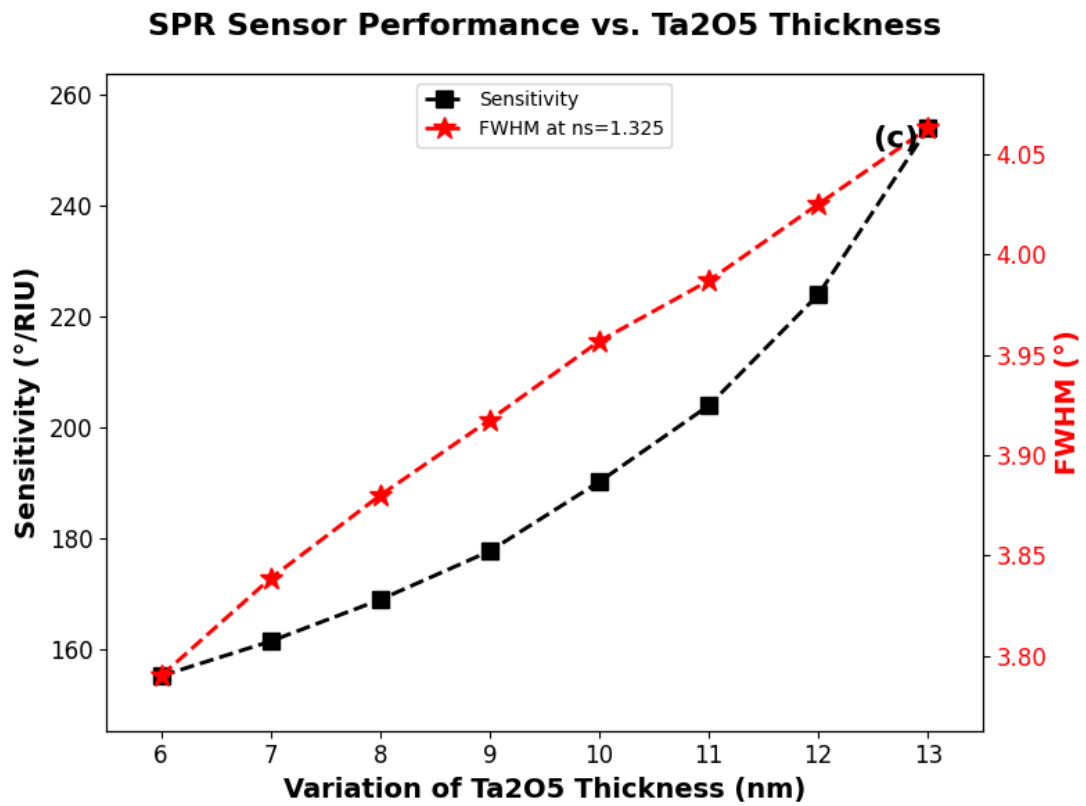


Figure 4.2: Optimization of BP layer thickness

Figure 4.3: Optimization of Ta₂O₅ layer thickness

4.2 Error Tolerance Analysis

The BP layer consistently exhibits the most robust performance, maintaining the lowest error rates even as tolerance increases. The Ag layer shows a moderate increase in error with tolerance. The Ta₂O₅ layer is significantly more sensitive to tolerance variations, displaying a substantially steeper increase in error percentage compared to the other two layers. This suggests that for applications requiring lower error rates, a lower tolerance setting is particularly crucial for the Ta₂O₅ layer, while the BP layer offers greater stability across a wider range of tolerance settings.

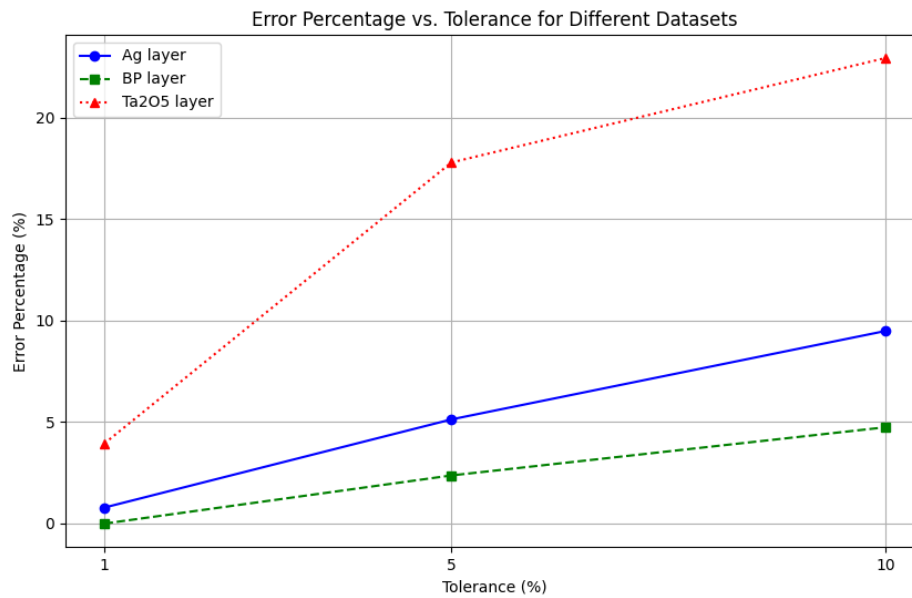


Figure 4.4: Error percentage vs. tolerance for different material layers (Ag, BP, and Ta₂O₅).

4.3 Melamine and Fat adulterants

Following the optimization of sensor thickness for detecting milk adulterants across a refractive index range of 1.325 to 1.365, this section specifically examines the implications for melamine and fat. These common adulterants are analyzed at various concentrations within milk to demonstrate the sensor's performance.

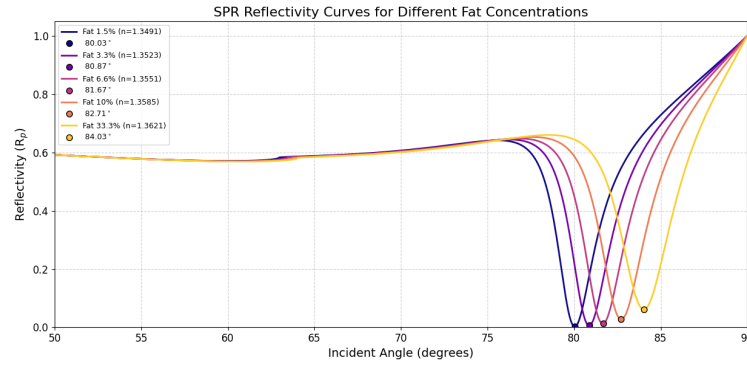


Figure 4.5: Reflectivity curve for different concentrations of fat

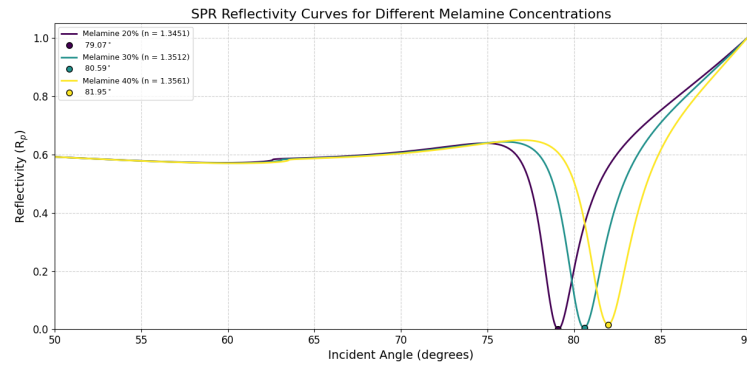


Figure 4.6: Reflectivity curve for different concentrations of melamine

Table 4.1: Optical parameters vs fat Concentration (1.5% - 33%)

Fat (%)	n	θ_R	S	FWHM	FoM	DA
1.5	1.3491	80.0303	234.3807	4.0444	57.9519	0.2473
3.3	1.3523	80.8710	246.8220	4.0884	60.3712	0.2446
6.6	1.3551	81.6720	257.6834	4.1404	62.2361	0.2415
10.0	1.3585	82.7130	269.8995	4.2204	63.9508	0.2369
33.3	1.3621	84.0340	290.3488	4.3204	67.2037	0.2315

Table 4.2: Optical parameters vs Melamine Concentration (20% - 40%)

Melamine (%)	n	θ_R	S	FWHM	FOM	DA
20.0	1.3451	79.0691	212.9789	4.0084	53.1331	0.2495
30.0	1.3512	80.5906	233.5669	4.0684	57.4099	0.2458
40.0	1.3561	81.9520	247.3811	4.1604	59.4607	0.2404

4.4 Comparison with existing techniques

The proposed **BK7-Cr-Ag-BP-Ta₂O₅** sensor configurations outperform existing works, particularly in terms of figure of merit (FOM) and detection accuracy (DA) within comparable refractive index (RI) ranges. While sensitivity is competitive and, in one instance, superior, the enhanced FoM and DA highlight the improved overall performance and practical applicability of the proposed sensor design for the specified RI ranges relevant to applications like milk adulterant detection. The balanced achievement of high sensitivity along with significantly better FOM and DA underscores the efficacy of the chosen material combination and structural optimization.

Table 4.3: Performance comparison of the proposed SPR sensor configuration with existing works.

Ref	RI Range	Configuration	S	FOM	DA
[6]	1.325-1.365	BK7-Cr Ag-Au-BP WS ₂ /sample	237.2	29.27	0.123
[7]	1.3791-1.3900	BK7-Ag-InN WS ₂ /sample	160.63	77.6	-
[4]	1.3404-1.3561	BK7-Ag-Pd WS ₂ / Sensing layer	248.96	45.8	0.184
Proposed Work	1.3404-1.3561	BK7-Cr-Ag-BP-Ta ₂ O ₅	231.30	56.67	0.2452
Proposed Work	1.345-1.3621	BK7-Cr-Ag-BP-Ta ₂ O ₅	259.82	62.34	0.24036

CHAPTER 5

CONCLUSION

The study successfully demonstrates the efficacy of a Surface Plasmon Resonance (SPR) sensor for detecting melamine and fat adulterants in milk. Through systematic optimization of the BK7–Cr–Ag–BP–Ta₂O₅ multilayer structure using the Transfer Matrix Method (TMM), the developed sensor achieves exceptional performance metrics, including high sensitivity (259.82°/RIU), precision, and robustness. The Kretschmann configuration enables real-time, label-free detection while maintaining operational simplicity, offering significant advantages over conventional analytical techniques in terms of speed, cost-effectiveness, and practical implementation. These findings validate the sensor's potential as a reliable solution for milk quality monitoring, addressing crucial food safety challenges posed by adulteration. The research establishes a foundation for SPR-based detection systems that can effectively safeguard public health through rapid identification of milk contaminants.

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